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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**
3. Select **Bridged** mode
4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SHH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SHH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtproto kernel_rdr proto kernel_rdr
        valid_lft 179068sec preferred_lft 83386sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:          0.0
Usage of /:           5.1% of 48.91GB
Memory usage:         5%
Swap usage:           0%
Processes:            139
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

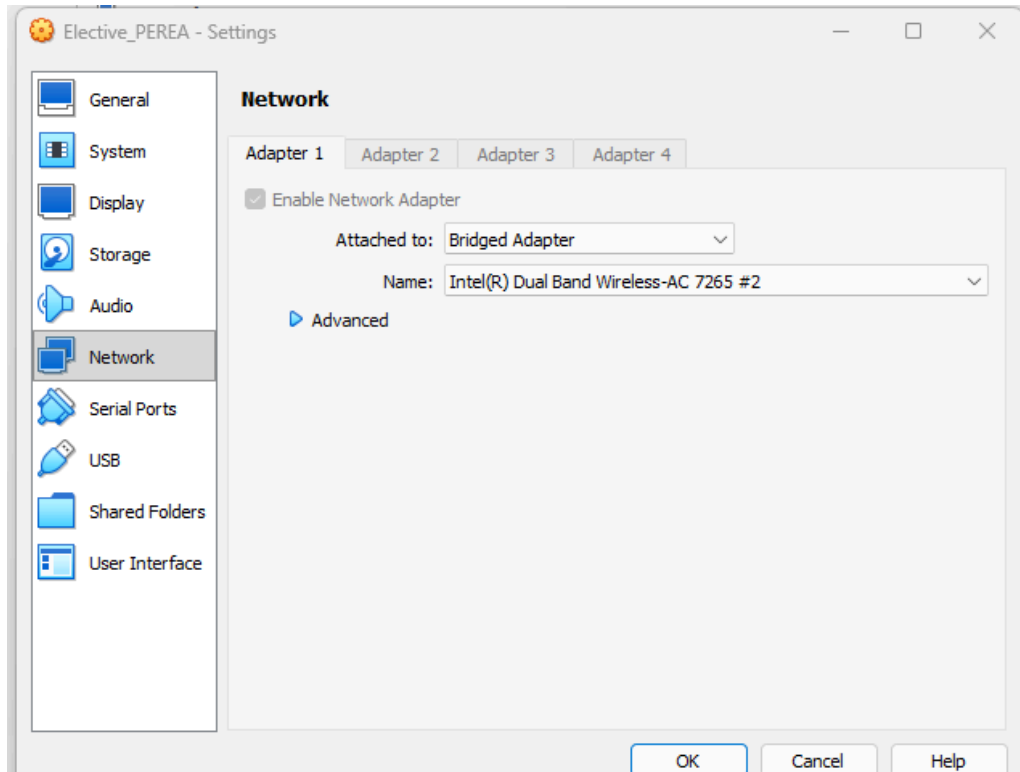
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter



- Server IP address (output of ip a or hostname -I)

```
server_perea@perea-virtualbox:~$ ssh server_perea@192.168.137.83
server_perea@192.168.137.83's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep  3 04:54:17 AM UTC 2026

System load:  0.1               Processes:            155
Usage of /:   14.5% of 47.93GB   Users logged in:     0
Memory usage: 8%               IPv4 address for enp0s3: 192.168.137.83
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu Sep  3 04:36:31 2026 from 192.168.137.58
server_perea@perea:~$
```

- SSH service status (sudo systemctl status ssh)

```
server_perea@perea:~$ sudo systemctl status ssh
sudo: 'systemctl': command not found
server_perea@perea:~$ sudo systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-03 04:01:44 UTC; 57min ago
   Invocation: a8109b44db1747089c076350fc4e38f4
   TriggeredBy: • ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Process: 1297 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1306 (sshd)
   Tasks: 1 (limit: 1567)
   Memory: 7.3M (peak: 9.2M)
   CPU: 120ms
   CGroup: /system.slice/ssh.service
           └─1306 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 03 04:35:59 perea sshd-session[2050]: pam_unix(sshd:auth): check pass; user unknown
Sep 03 04:36:02 perea sshd-session[2050]: Failed password for invalid user perea from 192.168.137.58 port 56358 ssh2
Sep 03 04:36:04 perea sshd-session[2050]: pam_unix(sshd:auth): check pass; user unknown
Sep 03 04:36:06 perea sshd-session[2050]: Failed password for invalid user perea from 192.168.137.58 port 56358 ssh2
Sep 03 04:36:06 perea sshd-session[2050]: Connection closed by invalid user perea 192.168.137.58 port 51804 [preauth]
Sep 03 04:36:06 perea sshd-session[2050]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.137.58
Sep 03 04:36:31 perea sshd-session[2053]: Accepted password for server_perea from 192.168.137.58 port 51804 ssh2
Sep 03 04:36:31 perea sshd-session[2053]: pam_unix(sshd:session): session opened for user server_perea(uid=1000) by server_perea(uid=0)
Sep 03 04:54:17 perea sshd-session[2593]: Accepted password for server_perea from 192.168.137.185 port 56023 ssh2
Sep 03 04:54:17 perea sshd-session[2593]: pam_unix(sshd:session): session opened for user server_perea(uid=1000) by server_perea(uid=0)
server_perea@perea:~$
```

- Successful SSH connection from host
- Verification commands executed inside VM

```

Sep 3 12:38
server_perea@perea: ~ — ssh_server_perea@192.168.137.83

perea@perea-VirtualBox:~$ ssh perea@192.168.137.83
The authenticity of host '192.168.137.83 (192.168.137.83)' can't be established.
ED25519 key fingerprint is: SHA256:qNMEfwidUXi2YnvMnfVunqxyuQwh8F10RbnDuX+yI
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:1: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.137.83' (ED25519) to the list of known hosts.
perea@192.168.137.83's password:
Permission denied, please try again.
perea@192.168.137.83's password:
dsPermission denied, please try again.
perea@192.168.137.83's password:
perea@192.168.137.83: Permission denied (publickey,password).
perea@perea-VirtualBox:~$ ssh server_perea@192.168.137.83
server_perea@192.168.137.83's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Thu Sep  3 04:36:31 AM UTC 2026

System load:  0.07               Processes:    153
Usage of /:   14.5% of 47.93GB   Users logged in: 0
Memory usage: 8%                IPv4 address for enp0s3: 192.168.137.83
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

server_perea@perea:~$

```

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1 OBJECTIVES (5 POINTS)

Check all that apply:

- ☐ Create and configure a Virtual Machine with Ubuntu Server
- ☐ Configure Bridged Network Adapter for network access
- ☐ Determine and verify the server's IP address
- ☐ Install OpenSSH Server and enable the service
- ☐ Establish successful SSH connection from host machine
- ☐ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Windows 11
Virtualization Software (VirtualBox/VMware)	Virtual Box
Ubuntu Server Version	26.4
VM Name	Elective_PEREA
RAM Allocated	4 GB
Storage Allocated	100 GB
Network Mode	NAT
SSH Client Used	VirtualBox

3 VM CONFIGURATION

Setting	Value
VM Name	server_perea
OS Type	Ubuntu (64-bit)

Memory	4000MB
Hard Disk	100GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed:

Server IP Address: 192.168.137.83

Network Interface: Physical

6. REMOTE CONNECTION

Connection Details:

Parameter	Value
Server IP	192.168.137.83
Username	server_perea
SSH Port	22